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# YOLOv9: Learning What You Want to Learn Using Programmable Gradient Information

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## Abstract

*Today's deep learning methods focus on how to design the most appropriate objective functions so that the prediction results of the model can be closest to the ground truth. Meanwhile, an appropriate architecture that can facilitate acquisition of enough information for prediction has to be designed. Existing methods ignore a fact that when input data undergoes layer-by-layer feature extraction and spatial transformation, large amount of information will be lost. This paper will delve into the important issues of data loss when data is transmitted through deep networks, namely information bottleneck and reversible functions. We proposed the concept of programmable gradient information (PGI) to cope with the various changes required by deep networks to achieve multiple objectives. PGI can provide complete input information for the target task to calculate objective function, so that reliable gradient information can be obtained to update network weights. In addition, a new lightweight network architecture – Generalized Efficient Layer Aggregation Network (GELAN), based on gradient path planning is designed. GELAN's architecture confirms that PGI has gained superior results on lightweight models. We verified the proposed GELAN and PGI on MS COCO dataset based object detection. The results show that GELAN only uses conventional convolution operators to achieve better parameter utilization than the state-of-the-art methods developed based on depth-wise convolution. PGI can be used for variety of models from lightweight to large. It can be used to obtain complete information, so that train-from-scratch models can achieve better results than state-of-the-art models pre-trained using large datasets, the comparison results are shown in Figure 1. The source codes are at: https://github.com/WongKinYiu/yolov9.*

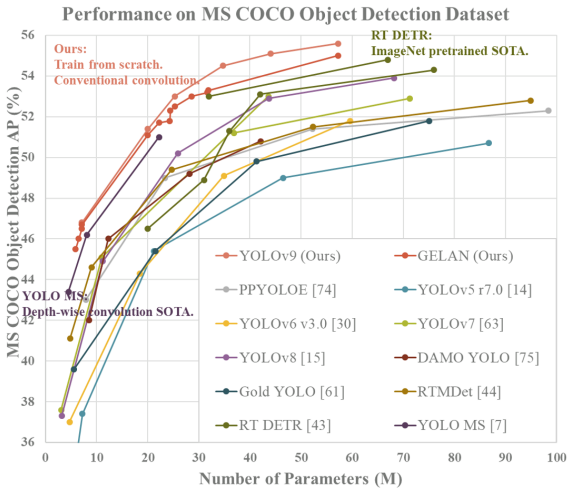


Figure 1. Comparisons of the real-time object detecors on MS COCO dataset. The GELAN and PGI-based object detection method surpassed all previous train-from-scratch methods in terms of object detection performance. In terms of accuracy, the new method outperforms RT DETR [43] pre-trained with a large dataset, and it also outperforms depth-wise convolution-based design YOLO MS [7] in terms of parameters utilization.

## 1. Introduction

Deep learning-based models have demonstrated far better performance than past artificial intelligence systems in various fields, such as computer vision, language processing, and speech recognition. In recent years, researchers in the field of deep learning have mainly focused on how to develop more powerful system architectures and learning methods, such as CNNs [21–23, 42, 55, 71, 72], Transformers [8, 9, 40, 41, 60, 69, 70], Perceivers [26, 26, 32, 52, 56, 81, 81], and Mambas [17, 38, 80]. In addition, some researchers have tried to develop more general objective functions, such as loss function [5, 45, 46, 50, 77, 78], label assignment [10, 12, 33, 67, 79] and auxiliary supervision [18, 20, 24, 28, 29, 51, 54, 68, 76]. The above studies all try to precisely find the mapping between input and target tasks. However, most past approaches have ignored that input data may have a non-negligible amount of information loss during the feedforward process. This loss of information can lead to biased gradient flows, which are subsequently used to update the model. The above problems can result in deep networks to establish incorrect associations between targets and inputs, causing the trained model to produce incorrect predictions.